PERFORMANCE ASSESSMENT OF A BILINGUALLY TRAINED MAXIMUM ENTROPY MARKOV MODEL WITH THE TAGLISH LANGUAGE

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Taglish is a very widespread predominantly spoken mixed language variety, whose phonology, morphology, syntax and semantics have been greatly influenced by English and Tagalog (Tangco & Nolasco, 2002). Taglish is significantly more complicated than simple borrowing and Taglicization of English terminology; it entails frequent, patterned, systematic alternation between the two languages, as well as blending, which is defined by the affixation of Tagalog grammatical elements to English lexical items and vice versa. The code-switching discourse has its own set of rules and regularities, and the morpheme, word, and phrase levels can all see alternations and insertions. Code switching is traditionally prevalent in spoken language but with the proliferation of social media such as Facebook, Instagram, and Twitter; it is becoming ubiquitous in written modalities and genres. can be seen at various language levels of representation for various phonological, morphological, lexical, syntactic, semantic, and discourse/pragmatic language pairs. It might happen within a sentence or across sentences (Alghamdi, 2019).

The context of English in the Philippines is established through fields such as science, health, education, and government because the default language used in these fields is English. The technical writing in these fields is almost exclusively in English. Moreover, street signs, advertisements, nutritional labeling, and place names are also in English (Lesada, 2017). Filipinos view English as a mean to attain a successful career, or an opportunity outside the Philippines. Sibayan (1972) expressed English as “the language of distance, of the semi-formal and formal”, yet some Filipinos see English as a colonial relic of the Philippines past, which is why some nationalists oppose it (Lesada, 2017). Tagalog on the other hand, remains as the dominant mother tongue of the

Philippines. Tagalog is also considered as a unique national identity whereas the creative writings in Tagalog seek to legitimize the Philippines' Identity as a true Asian country with its own language (Lesada, 2017). In contrast to how Sibayan (2017) described English, he expressed Tagalog as "the language of intimacy or nearness and remains the language of the home, the neighborhood, and the marketplace".

Filipino students are required by law to learn about two languages by the Policy on Bilingual Education (DepEd, 1987), which are most of the time, English and Filipino. The students are expected to graduate high school being fluent in English and Filipino because they took subjects related to them starting from grade 1. According to Lesada (2017), Filipinos do not hold fast to their mother tongue in the same way as westerners do. This notion is manifested in the quickness of Filipino to move from a mother a mother tongue to a local or national language and back. In terms of language attitude Filipinos are also quick to abandon one language to reap the social benefits of another. This was particularly evident in the Philippine's history where Filipinos adapted on speaking Spanish during the Spanish colonial era, and English during the United States colonial occupation. Another proof of this is the abundant number of loanwords that Tagalog speakers have incorporated from different languages. Filipinos are open to other languages when it comes to linguistic identity. Language choices evidently play a vital role in Philippines' society and personal expressions (Lesada, 2017).

Taglish is a result of societal diglossia and widespread bilingualism in the Philippines. According to Tupas (1998), code-switching is a manifestation of linguistic confusion and protest – a response to the centuries worth of shifting world views and philosophical stances as they pertain to language and identity. These kinds of sentiments show how deeply established language attitude and linguistic identity has become the very

foundation of modern Philippines Society. An appropriate conclusion for Taglish is that it is a product of history and complex language situations, juggling shifting language attitudes, policies, domains, and cultural phenomena. Taglish relies more on Tagalog than it does to English. It is heavily influenced by Tagalog grammar (Lesada, 2017).

Using Natural Language Processing (NLP), which is a field of study and application that investigates how computers can interpret and modify natural language text or speech to accomplish meaningful tasks, Languages can be studied further. NLP researchers want to learn how people interpret and use language so that appropriate tools and techniques may be developed to help computers understand and manipulate natural languages to execute their desired tasks. The foundations of NLP lie in several disciplines, viz. computer and information sciences, linguistics, mathematics, electrical and electronic engineering, artificial intelligence and robotics, psychology, etc. Applications of NLP include a few fields of studies, such as machine translation, natural language text processing and summarization, user interfaces, multilingual and cross language information retrieval (CLIR), speech recognition, artificial intelligence and expert systems, and so on (Chowdhury, 2003).

The benefits of the field of Natural Language Processing prove it to be useful as a human endeavor. With NLP, there are some top-level benefits that are of no denial even in a business setting (*Natural Language Processing (NLP): What Is It & How Does It Work,* 2019). It is possible to perform large-scale analysis of unstructured text data like social media comments, customer support tickets, online reviews, and news reports. Automating common processes in real-time without human intervention is also a possibility. The potential of Natural Language Processing is a thing to consider, where a tool exists that can be tailored to the user's needs.

Natural Language Processing research about Taglish are lacking even with its rich morphology. In Using Stanford Part-of-Speech (POS) Tagger for the Morphologically-rich Filipino Language (Go & Nocon, 2017), they included English nouns and labeled as foreign words and used an overwrite function of the Stanford POS Tagger to identify them. Qudah's (2019) work, which involved creating a constituency parser for the language, recommended that a POS tagger should be constructed and has many avenues to go to considering the resources present and the current progress in Natural Language Processing research and development.

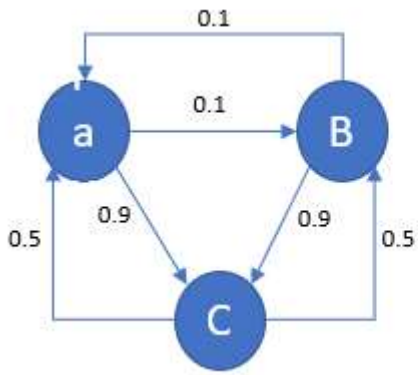
In this paper we present an attempt into POS tagging Taglish code-switched text, using a first order Hidden Markov Model (HMM) which will learn over different annotated monolingual corpora for both English and Tagalog. A maximum-entropy classifier will be utilized to tag unknown words as accurately as possible. A single POS tagger which understands extra-sentential, intra-sentential and inter-sentential code-switching between the two languages will be produced by the study. Finally, a manually annotated Taglish corpus validated by a linguist will be used for testing the validity of the model in the language. This will be accessible as a Python library and and as web application.

Theoretical Framework

This section presents the encompassing theories that support this study towards the creation of an integrated POS tagger for Taglish code-switched data.

The Theory of Markov Processes.

Figure 1. Example of a Markov Chain

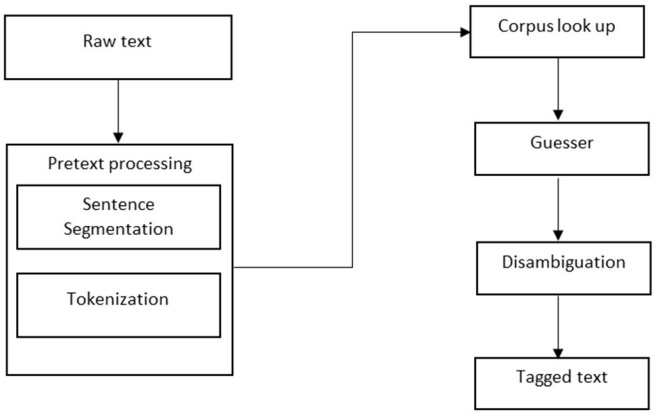


$$P = \begin{bmatrix} 0 & 0.1 & 0.9 \\ 0.1 & 0.0 & 0.9 \\ 0.5 & 0.5 & 0.0 \end{bmatrix}$$

A Markov chain is a model that describes the probability of sequences of random variables, or states, each of which may take on values from a given set. These sets may be words, tags, or symbols that indicate anything. It is useful when we need to compute a probability for a sequence of observable events. In many cases, however, the events we are interested in are hidden, we don't observe them directly according to Jurafsky and Martin (2021). The HMM allows to address both observable and unobserved events. The inputs and hidden events (such as POS tags) are causative elements in our probabilistic model. A keynote property of the Markov process is the so-called **Markov property**. It states that the next state of the system is only determined by the previous state it was in, and nothing else than that. This is an important process especially when modelling a POS tagger as a Markov chain with hidden states.

Part-of-Speech Tagging.

Figure 2. Architecture of a Part-of-Speech Tagger



According to Jurafsky and Martin (2021), POS is a useful clue to tell sentence structure and meaning. Knowing whether a word is a noun, or a verb tells us about likely neighboring words because in languages like English, nouns are preceded by determiners and adjectives. Knowing the POS can also be a hint on the syntactic structure of a text, making POS tagging, an integral aspect in parsing.

Tagging is a disambiguation task; words are ambiguous —have more than one possible POS —and the goal is to find the correct tag for the situation. (Jurafsky & Martin, 2021). For example, the word park can be a verb while it can also be a noun referring to a place.

The figure above shows the typical architecture for a POS tagger software tool shown by Zinsmeister (2010), which has several components where text passes through and gets transformed. Raw text to be annotated comes in the first step, where it is preprocessed and prepared by identifying the sentences, and the individual tokens. The guesser and disambiguation phase, depending on the model employed, receives the

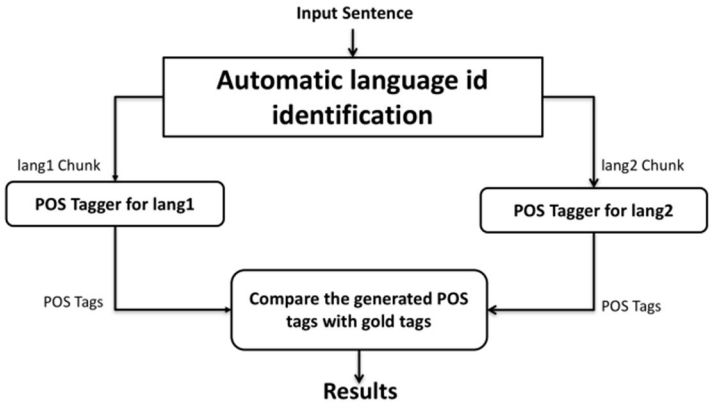
preprocessed text and annotates a tag for each token. The final output of the tagger is the annotated text and can be represented with any data structure such as a list or a tuple, or a combination of both.

A POS tagger can be formed in multiple ways as described in Chiche & Yitagesu's (2022) comparative analysis of different approaches, and is broadly categorized into two (2) types: the rule-based approach, being the oldest approach, is a rigorous methodology where expert linguists handcraft the rules and create a tagger or a parser out of it, and; the stochastic or data-driven approach, where statistical models are run into pre-annotated data such as a corpus, then the underlying structures are mathematically inferred by the algorithm.

Part-of-Speech Tagging for Code-Switched Languages.

POS tagging for code-switched languages can be approached in two (2) different ways, according to AlGhamdi, Molina, Solorio, Soto, & Hirschberg (2016). First is by using a combined framework where a monolingual POS tagger will be used for each language after the language has been identified. Then the integrated POS tags from both monolingual taggers to create the POS sequence for the input. The order can also be reversed where the POS tagging for each language will be done first, and then it will be identified. This approach can also be done with probability scores or with a machine learning framework like State Vector Machines.

Figure 3. Combined Architecture for Part-of-Speech Tagging Code-Switched Speech



The second approach is to use a single POS that exclusively annotates code-switched data, which is called the integrated approach. This can be done by training it with a dataset that has code-switched manually annotated data or by using the monolingual corpora for the languages, or even by using both code-switched data and monolingual data. This is the approach we will employ in the study.

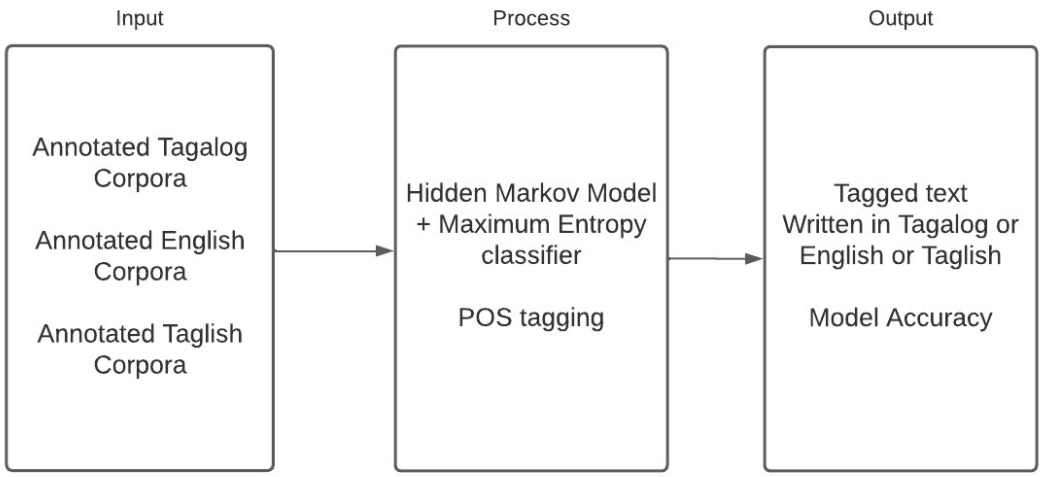
Depending on the language pair and the distance between them there are varying degrees of need for annotated code-switched data in the training phase of the process. Languages that share a significant amount of homographs when code switched will benefit from more code-switched data at training time, while languages that are farther apart such as Spanish and English, when code switched, benefit more from having larger monolingual data mixed (AlGhamdi et al, 2016).

Conceptual Framework

This section presents the conceptual framework of the study, which utilizes an IPO model. The model is broken down into three (3) distinct parts: the inputs, which are

modeled as consumables and efforts introduced into the study; the process, which crunches the inputs and performs operations around it, and the outputs, which includes generated tools and data of the study.

Figure 4. Conceptual Framework of the Study



The input would be annotated corpora, one each for Tagalog, English and Taglish. Processes include the construction and learning phase of a HMM, and the actual POS tagging when the model is given raw text. The output of the study would be the statistical model itself introduced as a POS tagger which will be hosted in a GitHub repository.

Statement of the Problem

The study aims to alleviate the need for a Taglish part-of-speech tagger as recommended in Qudah's work (2019), and to further endeavors in Natural Language Processing resources in the language. A POS tagger for Taglish code-switched data with acceptable performance is to be expected. Specifically, the study answers the following questions:

1. How does the model perform for each tag, with the following statistical measures?
    a. Precision
    b. Recall
    c. F Measure
2. How does the model perform overall, in terms of the following measures?
    a. Weighted Average Precision
    b. Weighted Average Recall
    c. Overall Accuracy
    d. Misclassification Rate
3. How does the model perform for the monolingual datasets used in the study, with the same measures in 1 and 2?

Scope and Limitations of the Study

This study is conducted as a partial requirement for the degree Bachelor of Science in Computer Science offered by the Polytechnic University of the Philippines. The research will be an experimental type, with the details specified in Chapter 3 Methodology.

Due to current circumstances of the world, the educational sector of the Philippines is limited in different factors. An online mode of learning is employed, and therefore the same mode is used even in both casual and important meetings held by the thesis group. Both communicability and availability of the students is vastly impeded by these circumstances and affect different parts of the study in general.

The datasets to be used for the study are 3 monolingual datasets, one (1) for English and two (2) for Tagalog. We will use the data from the Universal Dependencies Project (de Marneffe, Manning, Nivre, & Zeman, 2021) specifically UD TRG (Aquino,

2020) and UD Ugnayan (Samson, Zeman, & Tan, 2018) for Tagalog and UD GUM (Zeldes, 2017) for English. The contents of the Universal Dependencies Project (de Marneffe, et. al, 2021) are all available to download with licenses that will be expounded in chapter 3. We will then manually annotate the TweetTaglish dataset (Herrera, Aich, & Parde, 2022) then it will be used for Tagalog-English code-switched dataset. We will also be using the Universal Dependency tagset which is not based on the Wordnet tagset. These corpora will be expounded on chapter 3. The dataset mentioned will be used for training and testing our POS tagger.

The output of the study will be a Python package that can be deployed and published to the Python Package Index (PyPI), a repository of software. This package contains the trained model including an application programming interface specification for providing more data. Other than this, a web-based annotator tool application with visualization and summary will be deployed to the internet with a free hosting plan from the Heroku Platform-as-a-Service.

It will also not include other native dialects such as Cebuano and Bisaya. Even if these dialects have structural similarities to Tagalog, the dataset that we will use does not include those dialects which will require a multilingual approach. This can be used as a recommended future work instead, to further add resources to the Filipino Language.

Significance of the Study

Given the nature of the study from the creation of the natural language processing tool to the additional resources it yields, it aims to contribute to the following:

Natural Language Processing Resources. Code-switching is extremely used in most parts of the world where a country uses two or more languages (United Language

Group, 2022). Considering resources, popular social media sites such as Facebook and Twitter are filled with content from people that speak code-switched languages, especially Filipinos. Taglish code-switching is an important language to analyze, to further the progress into understanding it and the endeavors of Natural Language Processing in this instance.

Business Automation Tools. Businesses may be able to employ common automation tasks involving Natural Language Understanding such as named entity recognition for analyzing reviews or social media posts in Taglish. A POS tagger is an important early step in the creation of automated processes of this type.

Local and International. Filipinos could have a better understanding of their linguistic structure, which most locals are indifferent about (Thompson, 2003). Other local researchers could benefit from this research by using it as a reference or as a related study, which will help them further support their study. Additionally, a large portion of Social Media posts made by Filipinos can be syntactically analyzed with this research. Additionally, it could be used to show that Filipino is a high resource language that constantly evolves even in the sense of code-switching with English. Those who are foreign to the Filipino language, will benefit by better understanding the code-switching morphology (Lopez, Alexiadou, & Veenstra, 2017).

Definition of Terms

Annotated Corpus. A corpus that is grammatically tagged where the words are assigned a label based on their part of speech

Annotation. Linguistic information provided in a corpus.

Code-Switching. The language alternation that occurs when a text alternates between two or more languages in a single instance.

*Computational Linguistics. T*he branch of linguistics in which the techniques of computer science are applied to the analysis of language

Corpus. A collection of text organized into a dataset.

English. A West Germanic language of the Indo-European language family, originally spoken by the inhabitants of early medieval England. Considered as the most common language in the world

F-measure or F1 score. a measure of a model's accuracy on a dataset. It is used to evaluate binary classification systems

Hidden Markov Model (HMM). a statistical Markov model in which the system being modeled is assumed to be a Markov process with unobservable states

Natural Language Processing. A branch of Artificial Intelligence that is the manipulation of natural language.

Part-of-Speech Tagging. The process of assigning one of the parts of speech to the given word.

Precision. The number of positive class predictions that belong to the positive class.

Recall. The number of positive class predictions made from all positive examples

Stochastic Processes. A process involving the operation of chance.

Tagalog. An Austronesian language spoken in Luzon and neighboring islands and forming the basis of the standardized national language of the Philippines.

Taglish. The combination of Tagalog and English, both in name and in substance.

Tagset. A list of POS tags used to indicate the POS of each word in a corpus

Treebank. A syntactically processed corpus that contains annotations of natural language data at various linguistic levels

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter covers different literature related to the study. It will be used to help the researchers find the gaps in the field of interest and guidelines throughout the whole research study. This chapter also covers related studies and how they help the researchers in supporting the study.

Related Literature

Part of Speech Tagging.

POS tagging is the process of marking up a word in a text corresponding to a POS. The assignment of the word can be based on its definition or the context to its relationship with adjacent and related words in a phrase, sentence, or paragraph (Brill, 1992).

As POS tagging becomes a continually relevant task in Natural Language Processing. Different approaches to the task are plausible and are described in a comparative analysis paper presented by Chiche & Yitagesu (2022). The most familiar ones are described as follows:

- The **rule-based approach** relies on manually crafted rules to assign tags to words in a sentence. It may be constructed by linguistic experts, or roughly identify the rules using a machine-learning approach trained on an annotated corpus.
- The **artificial neural network approach** is a model inspired by biological neurons. The association between the neurons have numeric loads that are randomly generated at first, and then is changed through experience.

- The **maximum entropy model** is a discriminative model that predicts some class under certain conditions distributed in space. These conditions are the featureset, associated with learned weights.
- The **naïve-bayes approach** is a popular probabilistic model. It assumes that the features of data are independent of each other, and all contribute towards the probability of a label. It aims to calculate the posterior probability of all outcomes from prior data, and then chooses the outcome with the highest value.
- The **HMM** which is oversimply a probabilistic state machine. The states are hidden, and the model is given a series of observations. Implementors are interested with how the model reacts to its input.

In general, POS tagging is divided into three (3) broad categories. The oldest approach is the rule-based approach. The other approach is categorized as stochastic, where data is provided that is annotated with POS tags, and then running a statistical model which mathematically infers the structures. The last approach involves merging both a rule-based and stochastic approach, also known as a hybrid.

Code switched Languages.

Monolingual speech involves merely using a single language in communication and tagging such monolingual languages are already a common task. Code-switching speech, on the other hand, revolves around interchangeably using words and phrases from different languages as they suit the situation (United Language Group, 2022). Linguistic code-switching is mostly used within bilingual and multilingual communities, and the reasons range from the need to fit in a group, as a force of habit, or to convey thoughts and concepts that might be easier to explain in a specific language. Another definition of code-switching comes from a socio-linguistic perspective, where it describes the use of

different accents, dialects, language combinations and mannerisms within social groups in order to project a particular entity. With code-switching proven to be happening all around the world, the importance of analyzing code-switched languages continues to grow.

Figure 5. An instance of code-switching using Tagalog and English

'Pag nagsasalita ka, they'd say "Ay naku, she's trying to be better than thou".

POS tagging is already a delicate task to perform, especially creating models of any kind that automate it. Tagging text where code-switching happens further complicates this process, since such a piece of software must understand not only the individual languages on its own, but also instances where the languages in question are mixed in speech.

The Taglish Language.

The central language of focus in the study is the Taglish language, a code-switched informal language spoken in the Philippines, which involves both Tagalog and English. As a social phenomenon, it initially appeared in the late 1960s but gained traction following the 1973 Bilingual Education Policy; the biggest rise in the usage of the code-switching form occurred during the next fifteen years, from 1983 to 1988. According to Bonifacio Sibayan (1972), a Filipino linguist, claimed in 1985 that Taglish will become the nation's official language in the future. Code-switching Taglish is common among English-speaking Filipinos. A recent study suggests that young Filipinos can communicate successfully using Taglish but cannot speak competently in either language (Chureson, 2013). It may not be a problem in the Filipino Society because of how we successfully

understand each other using Taglish but it has a negative effect on communication between Filipinos and Foreigners as they cannot decode a Taglish message even it uses English in it. According to Lesada (2017), Taglish is far more complex than basic borrowing of English terms; it involves frequent, patterned, systematic alternation between the two languages, including blending characterized by the affixation of Tagalog grammatical elements to English lexical items and vice versa.

Taglish is primarily spoken in Metro Manila, but it has also spread to Filipino populations in other countries. It comes effortlessly to Filipinos who speak multiple languages, and it is easy to comprehend by other Filipinos. Advertisements, interviews, panel discussions, political speeches, and sportscasts here in the Philippines can be seen using the same code-switching variety of Tagalog and English. This is Taglish, or Tagalog-English code switching or Tagalog-English mix-mix, the alternation of Tagalog and English in the same discourse or conversation (Gumperz, 1982). it refers to the use of Tagalog words, phrases, clauses, and sentences in English discourse, or vice-versa.

There are 2 kinds of code-switching in Taglish: the first type of code-switching is deficiency-driven code switching whereas the person is not fully competent in the use of one language and therefore must go back to the other language to continue communicating. The second type of code switching is proficiency-driven code switching whereas the person is competent in the two languages and can easily switch from one to the other, for maximum efficiency or effect. Deficiency-driven code-switching can commonly be heard from kids who are still learning and trying to be competent in both languages since Filipino and English are a part of their studies. Proficiency-driven code-switching can mostly be noticed in professional interviews, where the host will switch languages for their audience to be more immersed in and for them to understand the

weight of the narrative more (Bautista, 2004). While according to the framework introduced by Hamers and Blanc (2000) and Poplack (1980), there are three main types in code-switching:

- **Inter-sentential:** switching happens at a clause or sentence boundary.

Figure 6. Instance of inter-sentential Taglish code-switching

Mommy, I don't want to answer this assignment. It's so hard, alam mo ba?

- **Extra-sentential:** switching of a single word or a tag phrase from one language to another.

Figure 7. Instance of extra-sentential code-switching

Bribery is such a ticklish subject. 'Pag nagsalita ka, huhusgahan ka nila.

- **Intra-sentential:** switching happens in the middle of a sentence.

Figure 8. Instance of intra-sentential code-switching

I won't pay attention to him kahit magkita pa kami on the street.

- And a special type of code switching that happens in the morphological level, otherwise known as **intra-word code-switching**.

Figure 9. Instances of intra-word code-switching

Nag-install, I-explain, Send-an, t-in-ext

Simple code-switching are just regular uses of Taglish in the Philippines while some words are morphed into having two languages together. These scenarios are called Mixed Verb Formation. According to Tangco and Nolasco (2002: 391), Tagalog is an

agglutinative language that features a complex verb system, using prefixes, infixes, and suffixes to mark aspect, mood, and focus.

Taglish Grammar and Structure.

Taglish is heavily influenced by Filipino more than English. Bautista (2004) stated that Taglish code-switching usually happens unconsciously because of our want for communicative efficiency. Because switching to the other language provides the most convenient way of expressing something through words with the least amount of effort

The evidence for this is the usual insertions of Taglish code-switching which are:

- **Function words** – adverbials that occur only in certain fixed word-order relations to other sentence elements and whose meanings constitute a rather heterogenous groupings.
- **Content words –** local words for local realities such as food words, kinship terms and culture-specific lexical items
- **Idioms –** metaphorical expressions that are available in one language but not available in the other.
- **Linguistic play** - achieving a humorous effect by playing on the Tagalog or English word.

This evidence indicates that those with proficiency in Tagalog and English has the strategic competence to measure what language would express what they are trying to say in the most concise way. Within the macro-view of Taglish, it is used for rapport, informality and solidarity and it is possible to look at certain instances of code-switching and explain them though a micro-view of communicative efficiency (Bautista, 2004). These uses constitute when does code-switching happen in Taglish in a Tagalog context.

Lesada's (2017) analysis in Taglish structure and grammar is done by interviewing Filipino professionals. It is stated that Tagalog is more used matrix language, or the more dominant language in a code-switching scenario, and English is the embedded language, or the language inserted into the grammatical frame of the matrix language. But it has also been proven that Filipinos can used Either Tagalog or English as a matrix language. Lesada (2017) analyzed the following structures:

Tagalog Enclitics.

One of the most basic code-switching phenomena in Taglish is the insertion of enclitic particles into English utterances. According to Moriguchi (2006), enclitics are a class of words and phrases that occur only in certain word-order relations to other sentence elements. These are important in conveying grammatical meaning. The clitics in the case of Taglish can be pronouns, interrogative markers, linking particles and phrases that show modality. These insertional switches are simply strategies of those who are prominent in both Tagalog and English for ease and lexical flexibility

Tagalog Conjunctions.

Another insertional practice is the use of Tagalog conjunctions in English frames. This will be the only insertion type in the study where English is a more dominant matrix language than Tagalog in a Taglish code-switching scenario. A valid question that comes to mind is why the bilingual would insert a Tagalog conjunction where an English one would function just fine. A possible answer is that such conjunctions display intonational patters that fulfill specific functions (Lesada, 2017). It is plausible that the intonations of the English conjunction could not convey enough what the speaker wants to express.

English content Morphemes.

Another observable form of Taglish code-switching. This is the insertion of English open-class lexical items like nouns, adjectives and adverbs into Tagalog Frames. Just like in inserting tagalog conjunctions in English frames, this is usually done to be succinctness and for communicative efficiency.

Embedded Languages Islands and Matrix Languages Islands.

This is based on the Matrix Language Frame by Myers-Scotton (1993). In a Taglish context, Tagalog speakers allow embedded language islands where the syntax of Tagalog and English are both compatible within their language boundaries. Taglish code-switching supports the Matrix Language Frame model and theory of Myers-Scotton (1993). Content morphemes from either Tagalog or English cannot be immediately considered as an embedded language island. Because the grammar of the matrix language might not allow a subject-verb order in the construction of a sentence, and it will not be right according to the grammar of the matrix language. A matrix language island, which in the case of Taglish is most of the time Tagalog, is comprised of the morphemes from the matrix language that are well formed because it follows it follows the grammar of the matrix language. The most similar form of an island connection is by connecting a matrix language island with an embedded language island with a dependent infinitive clause, which of course follows the grammar of the matrix language island. Though there are code-switching instances in Taglish that doesn't fit Myers-Scotton's proposed criteria.

Tagalog Tags.

Tagalog tags are content morphemes used to convey pragmatic meaning (Lesada, 2017). This is typically done when someone wants to express an interjection in the end of a conversational turn which is driven by the desire for social connection of a conversation expressed in linguistic means. An example of a Tagalog tag would be "*-char!*" which

people usually add on the end of a statement. Or “-ayun lang”. that indicates that you’re done expressing a statement.

Mixed Noun Phrases.

These are noun phrases that have a high frequency of alternation and insertion code-switching in a noun phrase. This does not fit the Myers-Scotton criteria because of the frequency of the code-switching makes it more complex. The richness of switches in a noun phrase is evidence of Taglish being proficient, flexible and creative as its own form of code-switching. It is apparent that the places where syntactic congruence exists between Tagalog and English allow for frequent alternation and insertion, resulting in a complex form of code switching that still follows the grammar rules.

Mixed Verb Phrases.

Tagalog is language that features a complex verb system with prefixes, infixes, and suffixes that provides more details on a verb. In Taglish, this is where intra-word code switching happens

Mixed Verb Formation has four types of affixing practices: prefixed, prefixed with reduplication, infixed and suffixed.

- **Prefixed scenarios** are Taglish words such as *na-move, i-naccept, nag-drive*. Adding Tagalog prefixes to English verbs serves to shift the focus or aspect of the verb for semantics.
- **Prefixed scenarios with reduplication** are Taglish words such as *na-haharvest*, *nag-swiswimming*, *nag-kocoffee*. This can happen in many cases in Taglish. This can mark a verb to have a progressive aspect. A more interesting use of the prefixes with reduplication is to use English nouns as

Tagalog verbs. This is done by adding Tagalog verbal prefixes and morphological reduplication to produce verbs out of English. This kind of practice is a part of Tagalog in nature (Lesada, 2017).

- **Infixed scenarios** are Taglish words such as *ki-no-mission, ki-non-sult, p-in-ickup.* This is used to convey a verb in its past tense, implying completion.
- **Suffixed scenarios** are Taglish words such as *fillup-an, drawing-an, recommend-an.* This is used to denote the verb for an object or a locative focus.

The use of affixes in Taglish might imply that it is simply just Using English words as borrowed words, but the evidences suggest that the verb formation of Taglish is indeed switching between codes (Lesada, 2017).

Resources in Computational Linguistics: Corpora and Treebanks.

Corpus is a term from the field of linguistics and refers to a large set of texts which is representative of a language and is used to analyze it. Corpora provide evidence of how a language is used in real situations and are a key resource for lexicography. A corpus can contain different number of languages, it can be monolingual, bilingual, or multilingual. Written texts or transcriptions of spoken language can be found in a corpus. It can contain literature from a single domain, such as medicine or agriculture, and it can collect texts from the same time or from several consecutive eras, or it can just be about the general language. An annotated corpus posits availability towards creating stochastic POS taggers.

According to McCarthy (2004), people create corpora of various sizes for various reasons. To assist in the preparation of a dictionary, for example, a big corpus would be required. Because it must include many examples of all the words and expressions used

in the language, it might contain tens of millions of words. A medium-sized corpus might comprise transcripts of lectures and seminars, and it could be utilized to develop books for students who need academic language. Corpora of this magnitude range from a million to five to ten million words. Other organizations are more specialized and smaller. These could, for example, contain the transcripts of business meetings and be used to assist writers in creating materials for teaching business language.

With corpora and software tools to analyze them, language can be seen on how it is really used. Intuition will no longer be in play to know what to read or write. With a corpus, it is easy to know what different people have said or written, which is easier than scanning a lot of documents for verification on how to use a language. It is not a theory of language learning or a teaching methodology, but it does influence our way of thinking about language and the kinds of texts and examples we use in language teaching (Mccarthy,2004).

An example of a large corpus is the British National Corpus, which is a 100-million-word collection of samples of written and spoken language from a wide range of sources, designed to represent a wide cross-section of British English from the later part of the 20th century, both spoken and written.

The written part of the BNC World (82%) includes, for example, extracts from regional and national newspapers, specialist periodicals and journals for all ages and interests, academic books and popular fiction, published and unpublished letters and memoranda, school and university essays, among many other kinds of text. The spoken part (17%) consists of orthographic transcriptions of unscripted informal conversations (recorded by volunteers selected from different age, region and social classes in a demographically balanced way) and spoken language collected in different contexts,

ranging from formal business or government meetings to radio shows and phone-ins (BNC Consortium, 2007).

Table 1. Composition of the BNC World Edition

Text type	Text	Kbytes	W-units	S-units	percent
Spoken demographic	153	4206058	4.30	610563	10.08
Spoken context-governed	757	6135671	6.28	428558	7.07
All Spoken	910	10341729	10.58	1039121	17.78
Written books and periodicals	2688	78580018	80.49	4403803	72.75
Written-to-be-spoken	35	1324480	1.35	120153	1.98
Written miscellaneous	421	7373707	7.55	490016	8.09
All Written	3144	87278205	89.39	5013972	82.82

A treebank is a linguistically annotated corpus with grammatical analysis that extends beyond the POS level. Although the most common way of representing a grammatical analysis is through the form of a tree, the term is not restricted to tree-shaped representations: it applies to all kinds of labelled corpora. There are a lot of factors in building a treebank since it is such a labor-intensive task like the availability of data and analysis tools The design of a treebank should be motivated by its intended usage, which can be for linguistics research or Natural Language Processing, yet most treebanks are desired to be designed to cater for both computational and corpus linguistics, with some overlapping goals.

An example of a treebank is the Penn Treebank. In its eight years of operation, (1989 – 1996), the Penn Treebank produced approximately 7 million words of POS tagged text, 3 million words of skeletally parsed text, over 2 million words of text parsed for predicate-argument structure, and 1.6 million words of transcribed spoken text annotated for speech disfluencies. The material annotated includes such wide-ranging genres as IBM computer manuals, nursing notes, Wall Street Journal articles, and transcribed telephone conversations, among others (Taylor, Marcus, & Santorini, 2003). It contains 36 POS tags and 12 other tags for punctuation and currency symbols.

Figure 10. The Penn Treebank POS Tagset

CC	Coordinating conj.	TO	infinitival *to*
CD	Cardinal number	UH	Interjection
DT	Determiner	VB	Verb, base form
EX	Existential there	VBD	Verb, past tense
FW	Foreign word	VBG	Verb, gerund/present pple
IN	Preposition	VBN	Verb, past participle
JJ	Adjective	VBP	Verb, non-3rd ps. sg. present
JJR	Adjective, comparative	VBZ	Verb, 3rd ps. sg. present
JJS	Adjective, superlative	WDT	Wh-determiner
LS	List item marker	WP	Wh-pronoun
MD	Modal	WP$	Possessive *wh*-pronoun
NN	Noun, singular or mass	WRB	Wh-adverb
NNS	Noun, plural	#	Pound sign
NNP	Proper noun, singular	$	Dollar sign
NNPS	Proper noun, plural	.	Sentence-final punctuation
PDT	Predeterminer	,	Comma
POS	Possessive ending	:	Colon, semi-colon
PRP	Personal pronoun	(	Left bracket character
PP$	Possessive pronoun	)	Right bracket character
RB	Adverb	"	Straight double quote
RBR	Adverb, comparative	‘	Left open single quote
RBS	Adverb, superlative	“	Left open double quote
RP	Particle	’	Right close single quote
SYM	Symbol	”	Right close double quote

The Penn Treebank also has a well-developed predicate-argument scheme to label each argument of the predicate with an appropriate semantic label to identify its role with respect to that predicate, as well as distinguishing the arguments of the predicate,

and adjuncts of the predication. The Penn Treebank first used skeletal context-free bracketing with limited empty categories and no indication of non-contiguous structures and dependencies (Taylor et al., 2003). during its release, but it was then changed to the Treebank II syntactic bracketing due to high demand for an annotation that would make explicit at least the logical subject and logical object of the verb, and indicate, at least in clear cases, how sub constituents are semantically related to their predicate.

Figure 11. Example of Penn Treebank II Skeletal Parsing

```
( (S (NP-SBJ-1 Jones)
(VP followed
(NP him)
(PP-DIR into
    (NP the front room))
    ,
    (S-ADV (NP-SBJ *-1)
            (VP closing
                (NP the door)
                (PP behind (NP him)))))
                .))
```

Some of the datasets to be used in the study are in the form of a flattened dependency treebank stored in a text file such as shown in Figure 11 without whitespaces. These treebanks also contain the POS on the word level which is the focus of the study

and where the model will learn lexical annotation probabilities, as elaborated in the following sections.

Machine Learning.

A machine learning algorithm is a computational process that uses input data to achieve a desired task without being literally programmed (i.e., “hard coded”) to produce a particular outcome. These algorithms are in a sense “soft coded” in that they automatically alter or adapt their architecture through repetition (i.e., experience) so that they become better and better at achieving the desired task (El Naqa and Murphy, 2015). Data samples and intended outputs are delivered together with the adaptation process, which is referred to as "training." In order to generalize to new, previously unknown data, the algorithm optimizes itself such that it can achieve the intended result when given the training inputs. The "learning" aspect of machine learning is this training. Initiation training doesn't have to be restricted to a short period of time at the beginning.

The ability to adapt through input from its environment is the main key of machine learning having dependencies from data. Machine learning serves to solve complex problems and patterns by analyzing the inputs, preprocess then learn the desired task. Machine learning enables the user to give a computer algorithm a vast quantity of data and have the computer evaluate and make data-driven suggestions and choices based on the input data alone.

A sample implementation of machine learning where a computer is programmed to play checkers. Two machine-learning procedures have been investigated in some detail using the game of checkers. Enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program. Furthermore, it can learn to do this in a

remarkably short period of time (8 or 10 hours of machine-playing time) when given only the rules of the game, a sense of direction, and a redundant and incomplete list of parameters which are thought to have something to do with the game, but whose correct signs and relative weights are unknown and unspecified. The principles of machine learning verified by these 'experiments are, of course, applicable to many other situations (Samuel, 2000).

Hidden Markov Model.

The HMM, being the model of concern for creation in the study, is a useful statistical tool for modeling generative sequences defined by an underlying process that generates an observable sequence (Jurafsky & Martin, 2021). HMMs have found use in many fields interested in signal processing, particularly voice processing, but they have also been successfully used to low level NLP tasks such as POS tagging, phrase chunking, and extracting target information from texts.

In POS tagging, it requires construction of a series of probability matrices on which it performs its guessing, which considers the output tags to be the states. A transition probability matrix is one such matrix which represents the probability that a model transitions to another state from a previous state. The structure of this matrix depends on how many previous states are to be considered, also known as the Markov chain's *order*. The other probability matrix is the emission matrix which represents all the words the model knows and the probabilities for each tag being assigned to a word. The two matrices are keys to performing the guess, which typically involves creating a trellis lattice and calculating the maximum a-posteriori path to determine the output tags of the model. An algorithm popular with probabilistic models such as this for performing guesses and calculating the path is called the Viterbi algorithm, which is used in the study.

Related Studies

Attempts to Tagalog Part-of-Speech Tagging.

This section presents the previous efforts made by both local and foreign entities into POS tagging, specifically with the Tagalog language, a basis language for Taglish.

TPOST: A Template-based n-Gram Part-of-Speech Tagger for Tagalog.

In the training, word features which include predefined words, affixes, upper case letters, and hyphens, and tagging rules are extracted which then yielded a 92% to 77% accuracy. The errors were attributed to stemming errors, cascading errors, lack of resources, simple scoring, lack of training, and ambiguity (Rabo, 2004).

MBPOST: Memory-Based Part-Of-Speech Tagger.

The training phase is the collection of sentence templates, predefined words (or lexicon) and feature-value pairs from the training corpus. The testing data is checked for words that are predefined or words that are in the lexicon. The features of the words that are not found in the lexicon or predefined words are matched with the feature-value pairs extracted. Initial testing of MBPOST showed a high accuracy of 64%. The low accuracy rate is attributed to some imperfections in the process followed such as the pointing system as well as to the small size of the corpus used

PTPOST4.1: Probabilistic Tagalog Part-of-Speech Tagger 4.1.

It implements the concepts of the HMM, the Viterbi algorithm, lexical and contextual probabilities. In the training phase, a nondeterministic finite state accepter is constructed such that every state is associated to a POS and the transitions show the possible sequences of POS occurring in the training data. Lexical probabilities, representing the possibility of a word being associated to a tag in every state, are derived.

Contextual probabilities representing the possibilities of the occurrence of tags depending on the n-1 previous tags are derived (Cortez, Navarro, Tan, & Victor, 2005). Using the nondeterministic finite state accepter and the associated probabilities and based on the theory of the HMM, the most probable sequence of tags corresponding to the read sequence of words is determined. The contextual probability depends on the context window size. It has an accuracy ranging from 91% to 97% without stemming

Tag-Alog: A rule-based Tagalog POS Tagger.

For its training, a database of word-tag pairs and a patch list are derived. During testing, an initial tag is derived based on the word-tag pairs and the tag is refined using the patch list. Tag-Alog performed with an average accuracy of 78% on two corpus domains (Fontanilla & Wu, 2006). The errors were attributed to the tagging errors during the manual preparation and to the great number of unknown words.

To analyze these Old Tagalog POS taggers, they had to acknowledge that each POS tagger has their own proper format which requires the meticulous preparation of data that they will use for testing and training. They developed utility programs to facilitate and analyze the data such as: text file formatters, word-tag matchers, and token counters. The following processes were then performed:

- Implementing the system such that learning by applying the training data pairs (text and tag files) is performed
- Manually recording problems and other pertinent observations encountered during the training
- inputting the text file of words to be tagged
- recording tagging results and execution time

- performing analyses with the aid of text-based word-tag matching program, counter program and a worksheet application program

The results are as follows:

Table 2. Comparison of Tagging Results on Mutually Exclusive Testing and Training Data (Miguel & Roxas, 2007)

POS tagger	Accuracy (training data)	Accuracy (testing data)
TPOST	70%	65.2%
MBPOST	77%	85%
PTPOST4.1(without stemming)	78.3%	73.5%
Tag-Alog	72.5%	61.1%

These test results show that PTPOST4.1 without stemming and MBPOST consistently performed better than TPOST and Tag-Alog, whether the testing data was taken from the training data, or not. But it is significant to note that the approaches used by PTPOST4.1 without stemming and Tag-Alog may perform better in tagging over MBPOST and TPOST, respectively, even if the tagged corpus for the natural language under consideration is not significantly large, since the performances of the approaches of MBPOST and TPOST are dependent on the breadth of the corpus used for training (Miguel & Roxas, 2007)

SMTPOST: Using Statistical Machine Translation Approach in Filipino Part-of-Speech Tagging.

Statistical Machine Translation (SMT) is a translation technique which uses statistical models as its heuristics. By setting a parallel corpus as training input, it determines the patterns and matches of both words and phrases, together with their probabilities. SMT as a tagger is uncommon; as specified in its name, it is mainly used in

translating one language to another. However, it is not limited to being used that way. As a data-driven approach, the method for this research leverages SMT by using pairs of word features and POS tag counterparts and translated Filipino Wikipedia data as input for training; while for POS tagging, words or sentences are accepted as input to be automatically transformed into features to match the generated model from training. It got an accuracy of 84.75% but it still has its own set of advantages and disadvantages (Nocon & Borra, 2016). One of the advantages is related to its tagging process, which makes use of generalized data instead of literal ones. Choosing this type of data extends the tagger's scope and lessens the instances of out-of-vocabulary words. Applying statistics as basis is equally as important, for it uses frequency and probability to determine the correct tags, even for phrases and ambiguous words. On the other hand, disadvantages include a weak feature extractor and the lack of training data, hindering SMTPOST to tag complex feature combinations.

SVPOST: Part of Speech Tagging for Tagalog using Support Vector Machines.

Highlighting with the work of students at University of the Philippines Diliman - Computer Vision and Machine Intelligence Group, Dept. of Computer Science, they explore the use of Support Vector Machines and bi-grams in the POS Tagging of Filipino words through the creation of SVPOST. Introducing Support Vector Machines which are a set of kernel-based supervised learning methods mainly used for classification and regression analysis which analyze data and recognize patterns (Reyes, Suba, & Naval, 2011). By feeding it with data consisting of a corpus tagged words composed of unique words, symbols, and a tag set then SVPOST will utilizes the use of bigram patterns.

In their simulations using different kernel variations with different parameters, the radial kernel with a gamma of .5 was the one that produced the best result with an accuracy of 79.50%. The use of SVMs offers a promising approach in POS tagging giving as high of 81% additional with using the radial kernel even on longer build time of tags and larger gamma parameter will produce more accurate results.

SPOST: Using Stanford Part-of-Speech Tagger for the Morphologically Rich Filipino.

The key highlight of this POS tagger presented by is using the already-existing Stanford Part-of-Speech Tagger (SPOST) but with the Filipino language (Go & Nocon, 2017). The tagger was originally written by Kristina Toutanova in 2003 and is now maintained by the Stanford Natural Language Processing Group, and it is the highest-performing POS Tagger usable for multiple languages. Its high accuracy is accompanied by a Maximum Entropy Cyclic Dependency Network algorithm at its core.

A Filipino corpus was built containing 15,166 sentences with a total of 406,509 tokens that were manually translated Wikipedia pages and tagged with POS tags by Filipino linguists, as part of the stakeholders of the project, using the MGNN tag-set introduced in the SMTPOST. The research also considered different Filipino linguistic phenomena and select events of intra-sentence code-switching, training SPOST with 80% of the corpus and the remaining 20% was used for testing. A high accuracy of 96.15% was achieved with this corpus, which is larger in comparison to the previous attempts at tagging.

CRFPOST: Part-of-Speech Tagger for Filipino Texts using Conditional Random Fields.

Another attempt at Filipino POS was presented by Olivo, Hari, & dela Fuente (2019) using a CRF model. The Conditional Random Fields model takes neighbors into account when predicting a label. The attempt utilized the same corpus in the previous POS tagging attempts like SPOST, and the CRF++ Toolkit for developing the model. This attempt yielded an accuracy of 90.59%, which is considerably high and is like the results achieved by HPOST and SMTPOST.

Part-of-Speech Tagging Bilingual Speech Transcripts with Intra-sentential Model-Switching.

A study made by Rodrigues and Kubler (2013) presents a series of attempts to POS tagging code-switched text, particularly with the Spanish and English languages. The study first establishes baseline accuracies by (1) training on the dominant language corpus then testing on the other language corpus which yielded 24.11% and a mixed-language corpus which yielded 77.27% accuracy. The next baseline attempts (2,3) were to create a single model trained with both languages, whilst considering a consistent tagset. They present two approaches to this: an attempt where the tagsets for the individual languages are unmodified and merged which yielded an overall accuracy of 89.96%, and a language-distinct tagset where each tag is prefixed with the language (i.e., LANGUAGE-NAME_OF_TAG) and overlaps are replaced with a unique tag which yielded an overall accuracy of 89.50%. The final attempt (4) used two tagging models and relied on a language identification function which facilitated control over which model guesses a tag for a word, on which the weighted average reached 90.45%.

A Maximum Entropy Model for Part-Of-Speech Tagging.

Ratnaparkhi (1996) utilized a maximum entropy model that simultaneously uses many contextual features to predict POS tags. He used the data from the Penn Treebank

(Taylot et. al, 2003) to conduct an experiment with the model. For the model to work, a set of features is needed that will check the features of every token in the corpus, and if a token from the training set has a feature that exists within the predefined set of features, the model parameter will contribute to the joint probability of the feature and the corresponding tag. This joint probability will be used to make predictions on tagging unknown words. This relies on the hypothesized distinction that rare words in the training set are similar to unknown words in the data. The model yielded a 96.63% total word accuracy and an 85.56% unknown word accuracy which shows that the model performed well.

Parsing Code-Switched Taglish Language by Creating Constituents.

This study made by Qudah (2019) presents a way of creating a constituency parser for the Taglish language, which creates a constituency tree out of a POS annotated text. A constituent represents a group of words that act together as a single unit, and it can be nested within another. The goal is to generate constituents of parsed code-switched Taglish Language from the tweets in Twitter. The data was manually annotated by judges then the frequency of bigrams was checked to have a standard on the acceptable bigram's combinations, which performed with 41.7% accuracy. While a constituency parser is out of scope of the study, Qudah recommends that a POS tagger for the language be created to further endeavors in studying Taglish, since they manually annotated their data in a preliminary phase before the annotation of syntactic structures.

Different Approaches to Unknown Words in a Hidden Markov Model Part-of-Speech Tagger.

A problem with HMM is the occurrence of an unknown word in the text being processed. If a word is not in the training text, then the emission probability of the word

will be equal to zero (0), and the complication with this that all possible state-sequences containing an unknown word, will have the probability of zero (0) and therefore cannot be tagged with any annotation. It will also not be able to proceed because the HMM alone will not be able to transition to the next state with the transition probability matrix being 0 for every given tag. This problem will have a significant effect to the accuracy of the model. Haulrich (2009) presented some ways on solving this problem:

Laplace Smoothing.

Using this smoothing method, we add some of the probabilities in unforeseen events. To avoid estimating any probabilities to be zero for the events never observed, one (1) will be added to all frequency counts to pretend that the token has been seen once and then re-normalize the probability space.

Most probable POS-tag.

The method here is to find the most frequent POS-tag and assume that the unknown words will always have the said tag. This implies that the emission probability of all unknown words the said tag will be automatically one (1) and the emission probability for the other tags will completely be zero (0).

Overall POS distribution.

To avoid the flaw of distributing all the unknown words being distributed to the most probable POS tag, another method is to check the overall distribution of the POS tags for the known words and assume that the probability of an unknown word having a specific tag is the same probability of a known word having that specific tag,

Open word-classes POS distribution.

Some world classes are closed and adding new words to them cannot be done easily because it consists of words that serve as key grammatical functions (e.g pronouns, prepositions, and determiners). While open word classes are classes that are open to the addition of new words. An assumption that we can add given this information, is that the probability of an unknown word being tagged into an open world class is greater than the probability of an unknown word being tagged into a closed word-classes. Though if a word from a closed word class is no present in the training set, it will never be tagged correctly.

Hapax legomena.

In this method, the distribution of POS tags to unknown words is similar to the distribution of words that only occurred once in the training set. This assumption will be added to the distribution of unknown words.

Hapax legomena in open word-classes.

A combination of two (2) of the methods stated above, this we assume that the probability of an unknown word being tagged into an open world class is greater than the probability of an unknown word being tagged into a closed word-classes and the distribution of POS tags to unknown words is like the distribution of words that only occurred once in the training set.

Maximum-entropy classification.

A maximum-entropy model was used for the estimation of the emission probabilities. A set of features was used to classify the unknown word and the model was able to classify the most probable tag for a word, which is then treated as its emission probability. This will be the method that we will use and is expounded in Chapter 3 Instrumentation.

Synthesis of the Reviewed Literature and Studies

With the current state of linguistic resources for Taglish, the study aims to improve resources in the language which has established roots with Filipino POS tagging attempts. No part-of-speech tagger for Taglish currently exists. Tagalog language resources, especially POS taggers, were continually created and tested by different researchers from popular Philippine universities and is an important starting point into providing Taglish NLP resources. The Stanford Part-of-Speech Tagger for the Morphologically rich Filipino language by Go and Nocon (2017) have accounted for intra-sentential code-switching phenomena with English. We build above this by creating an integrated model trained with monolingual datasets of the two languages, which should account not only Tagalog and English on their own, but also any code-switching phenomena of the two languages.

The basis of grammar for Taglish depends on its user. But the more prominent behavior of Filipinos is to use Tagalog as the matrix language for code-switching. So, this study will focus on Taglish with Tagalog being a matrix language and English being its embedded language in code-switching. Taglish Code-switching using English as the matrix language and tagalog as the Embedded language is still prominent, as Filipinos are proven to have the bilingual competence to use Both languages as a matrix language. Taglish code switching can happen in many instances and levels, reaching as deep as code-switching between morphemes going deeper than word-level code-switching, nevertheless it still follows the grammar rules of Tagalog which is an analysis of Lesada (2017).

The Universal Dependencies project (de Marneffe, 2021) is of large importance in the study. This presents a single universal Natural Language framework for different

languages with a set of details for part-of-speech tagging and dependency tree structures. The tagset concocted for the UD project is called the Universal POS Tags, which the publicly available dataset of the project uses for annotating the parts-of-speech of the individual words. The study will make use of three treebanks from this project, particularly UD GUM Treebank (Zeldes, 2017) as the English monolingual resource, and both UD TRG (Samson et. al, 2018) and UD Ugnayan (Aquino, 2020) merged as the Tagalog monolingual resource.

There are some works considerable as contributions in NLP for the Taglish language, one such as done by Qudah (2019) which focuses on creating a constituency parser. As a preprocessing phase, the study performed manual POS tagging as assisted by fluent speakers of the language. Qudah suggests that the annotation task should be automated with a tool and says that it is a highly viable option with different avenues.

The keynote studies that will assist the creation of a Taglish POS tagger is one performed by AlGhamdi et al. (2016) and a Spanish-English series of attempts made by Rodrigues and Kubler (2013). They analyzed different approaches to doing the task with a code-switched language. This presents multiple avenues the Taglish POS tagger can proceed to, from taking advantage of monolingual datasets, using an annotated code-switched dataset, to different POS tagger architectures that can analyze both languages in these scenarios.

Given the simplicity of Hidden Markov Models, it will be the model of choice for implementing a first part-of-speech tagger for this language. A helpful property of this model is the Markov property wherein a state relies on the previous state and is greatly used in modelling part-of-speech tag sequences. However, the large concern with probabilistic modelling is when an event is not observed at all in the data which results to

zero probability; this is a problematic occurrence since if at a certain word the model encounters these zero probabilities, then the next sequences cannot be guessed anymore. To add to this problem, Taglish speech contains code-switching even in the word level, where morphemes come from different languages such as *nagsya-shopping*; these words are never present in the monolingual datasets that the model will train from. In alleviating this problem, the study follows Haulrich's work (2009); a maximum entropy classifier will be trained with the same corpora which comes with an engineered featureset to account for these phenomena. The overall tagger architecture including this featureset is elaborated in Chapter 3 Instrumentation.

Chapter 3

METHODOLOGY

The chapter presents the research design, data sources, instrumentation, gathering procedures, analysis and collection techniques to be employed in performing the research.

Research Design

In performing the research, the researchers will use the **experimental** research design. To be precise, the **Post-Test Only Control Group Design** experimental type of research will be employed, which has four (4) characteristics (Posttest-Only Control Group Design, 2018) as elaborated in the following:

a. Two (2) or more groups are formed: a group that will receive intervention, called the intervention group, and a group that will remain stale, called the control group.
b. The treatment or intervention is administered.
c. Data is collected after the treatment is administered.
d. Data are compared between the two groups to determine if the intervention was effective.

Three (3) corpora will serve as the dataset for the model to be constructed, as elaborated in Chapter 3 Sources of Data. These corpora will be divided into two groups, a **training set,** and a **testing set**, which will serve for analysis later as elaborated in this chapter's Data Analysis section. The groups themselves won't receive intervention but rather is fed to the model and determine how it would perform in terms of F1 statistic of each tag. True randomization is used upon dividing the dataset into the two groups.

Sources of Data

The study aims to create an integrated single-model POS tagger that is capable of understanding Tagalog-English code-switching phenomena. To do that, a set of two different annotated corpora for English (1) and Tagalog (2).

The first two (2) datasets will be monolingual resources for both English and Tagalog respectively, both contributed to the Universal Dependencies project (de Marneffe, 2021). This project is of large importance since it has proven the effectivity of a single universal tagset not only in the two languages to be considered in the study but also with other languages, which is a collection known as the Universal POS Tags. The English resource will be a treebank called the UD GUM Treebank (Zeldes, 2017) which is a dataset consisting of 162,319 tokens with 9,130 sentences with parts of speech and dependency links. The Tagalog resource will be the two available treebanks known as UD-TRG and UD-Ugnayan which in total comprises of 1,745 tokens with 222 sentences, with the same nature as the previous English treebank.

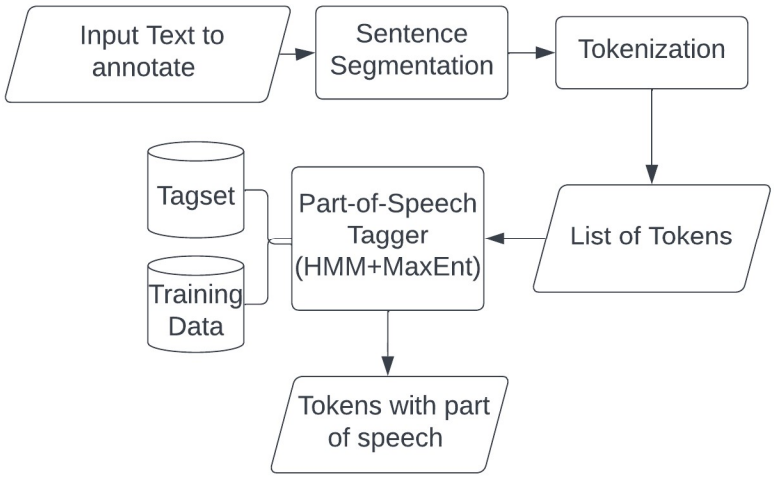
As an important dataset for testing the validity of the model in the language in question, the TweetTaglish corpus presented by Herrera et al. (2022) will be used which is unannotated, and the files only contain tweet id references. A golden standard dataset will be constructed from this corpus, to be manually annotated by the researchers and then validated by an expert linguist. The dataset will also focus on Tagalog being the Matrix language and English as the embedded language. The data will be annotated with the grammar and structure analyzed by Bautista (2004) and Lesada (2017) on Taglish. The tagset that will be used is the Universal POS tags utilized in the Universal Dependencies Project (de Marneffe et. al, 2021).

The study aims to determine the performance of the model not only with the Taglish language, but also with Tagalog and English by itself for each. Going further, a testing set for the three (3) languages must be present, where the Taglish resource is elaborated beforehand. For all languages, a 20% testing dataset will be extracted for each language resource. In this regard, the **simple random sampling method** will be employed where each sentence in the treebanks have an equal probability of getting grouped into either the training or testing set. The Universal Dependencies project has already separated each language resource into a training and testing set which eases the task.

Instrumentation

System Architecture and Tool.

Figure 3. The Taglish HMM Part-of-Speech Tagger

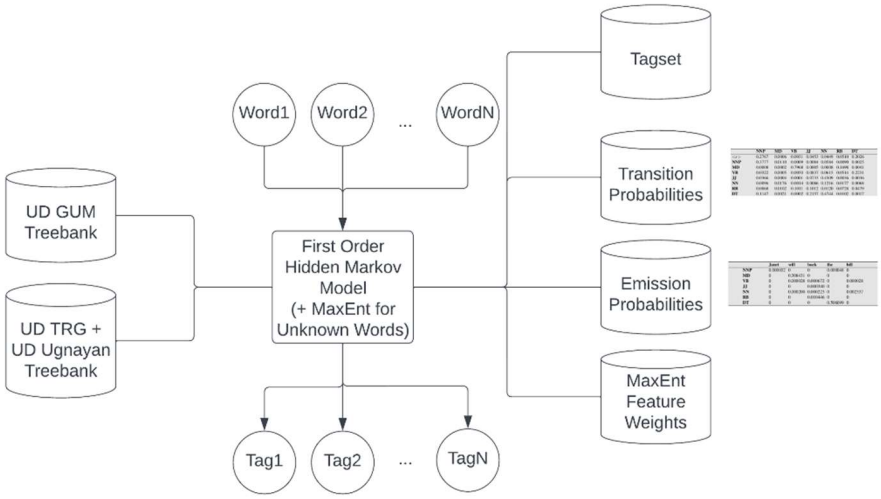


The POS tagger tool generated by the study will be constructed as a combination of components that interact with each other, as shown above.

The first step in any natural language processing task, not differing from the tool this study generates, is to preprocess the text to prepare it for further steps in the system. This process consists of a sentence segmentation process, which determines the boundaries of each sentence while considering different scenarios such as usage of punctuation marks, and a tokenization module, which separates the whole sentence string into a list of individual words or tokens.

The next part of the process is the POS tagging stage, where the sentences and its tokens are annotated with POS tags. The output generated by the previous phase, a list of tokens, will be plugged into the model. This tagger is an HMM, a statistical model. The phase outputs a list of 2-tuples for each token and its associated tag.

Figure 4. The Integrated First Order HMM + MaxEnt Architecture



A first order HMM will be implemented as the POS tagger for the study. States of the model will consist of all the tags in the tagset section which amounts to 17 in total. The input of the model will be a sentence with its tokens separated as a list of strings e.g.

[“This”, “is”, “maganda”], and the output will be a list of 2-tuples where each word has been associated a POS tag e.g. [(“This”, “DET”), (“is”, “VERB”), (“maganda”, “ADJ”)].

The figure above shows that the model will learn its parameters from three training sets drawn on the left side, as described in Chapter 3 Sources of Data. From here, the transition and emission probabilities derived from the corpora with the maximum likelihood estimation are saved into memory, which can also be persisted into a file, so the model does not need to be re-trained.

Tagging Unknown Words with a Maximum Entropy Classifier.

However, unencountered words in the corpora posits zero emission probabilities, which is extremely problematic in sequence labelling. To solve this, we also follow Jurafsky and Martin’s discussion (2021) into creating a **Maximum Entropy model** for classifying unknown words, which is also done by Haulrich (2009) as an addition to a baseline Hidden Markov Model. The features to be considered are as follows:

- $f_1 =$ Word contains a certain prefix
- $f_2 =$ Word contains a certain suffix
- $f_3 =$ Word contains an uppercase letter
- $f_4 =$ Word contains a hyphen
- $f_5 =$ Word is all in uppercase

Each feature in this set spans both the languages in question—Tagalog and English. Weights are associated for each feature, which is learned with a convex optimization algorithm. These weights are the parameters of the maximum entropy classifier; they can also be persisted into a file to prevent re-training of the model when shutting down the program.

Tagset.

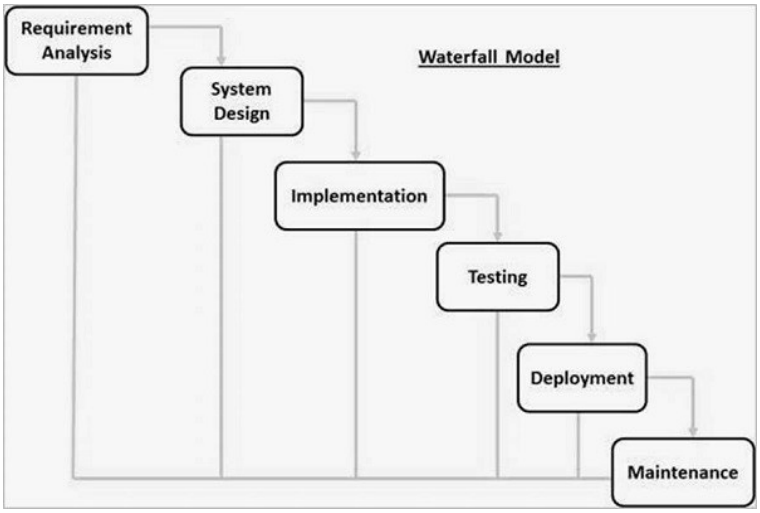
Figure 5. Listing of the Universal POS Tags, Universal Dependencies Project (de Marneffe, 2021)

- ADJ: adjective
- ADP: adposition
- ADV: adverb
- AUX: auxiliary
- CCONJ: coordinating conjunction
- DET: determiner
- INTJ: interjection
- NOUN: noun
- NUM: numeral
- PART: particle
- PRON: pronoun
- PROPN: proper noun
- PUNCT: punctuation
- SCONJ: subordinating conjunction
- SYM: symbol
- VERB: verb
- X: other

The tagset to be used, which will serve as the output of this classifier model, is the Universal POS Tags utilized by the Universal Dependencies project (de Marneffe, 2021). This Natural Language framework is particularly useful since it provides universal baseline features for any language. It consists primarily of parts of speech that are recognizable for the languages supported in the project, including Tagalog and English.

Development Details.

Figure 6. The employed Waterfall Model for the study



Researchers of the study will use the **waterfall model** where the whole project is separated into large phases from requirements analysis to the implementation of necessary features. Requirements analysis and system design is performed in the first half of the whole study, and the gathering of data, implementation of the model, and finally testing procedures are done in the second half of the study, the details of which are elaborated in the following sections. Considering difficulties in an online setup, an average of 2 hours every day will be spent contributing to the progress of the project until its completion.

The requirements Analysis phase consists of the creation of the idea, the formulation of the problem statement and the process of determining all the requirements necessary to perform the study. In this case, the POS tagger and its necessity and significance is determined, and the specifications for developing the model including the tool associated with it is formally defined. The subsequent step is the System Design, where the model's system architecture and the specification of the sources of data,

calculations, and analysis to be performed is defined. All these steps are performed in the subject Thesis Writing 1.

The implementation and testing phase consists of important steps in the production of the model and tool. The implementation phase elaborates on the plausible development of the model as defined in the specifications, and the training of the model's internal probabilities. The application tool that utilizes this tagger will also be developed in the span of this phase. The testing phase will focus on extracting statistical measures in evaluating the performance of the model, as defined in Chapter 3 Data Analysis.

As part of the last two phases, the source code of the model including a trained model that is serialized into a file and all the datasets will be publicly hosted in GitHub, in accordance with the Data Privacy Act for the corpora utilized. The source code repository will bear the MIT License.

Development Tools.

Python 3.10. A general-purpose computer programming language with dynamic semantics. This will be used to implement the different tools necessary in making our Taglish POS tagger.

Django Framework. A high-level Python web framework for rapid development. This will be used to implement the tool's website.

Twitter Application Programming Interface (API) v2. An API that enables programmatic access to Twitter in advanced ways. This will be used to access the TweetTaglish dataset by Herrera et. al (2022).

Natural Language Toolkit (NLTK). A leading platform that used for building Python programs for NLP. This will be used for its text processing abilities (i.e tokenization and tagging), and the Max-Ent Classifier model.

Research Instrument.

Researchers will use the following experiment papers to assess the relevance of using the integrated POS tagger. Each paper is elaborated in the following sections.

Table 3. Experiment Paper for the HMM Integrated Taglish POS Tagger confusion matrix for each language resource

		Predicted Tag			
Actual Tag		Tag 1	Tag 2	…	Tag N
	Tag 1				
	Tag 2				
	…				
	Tag N				

Table 3 above shows the unaggregated performance of the constructed POS tagger by tallying each tag frequency in a confusion matrix. This matrix will show when a tag was properly predicted and if not, what the predicted tag was in that case. Subsequent measures will be built upon this experiment paper. Tags in each dimension will be the Universal POS Tags, as elaborated in Chapter 3 Instrumentation.

Table 4. Experiment Paper for the HMM Integrated Taglish POS Tagger performance matrix for each tag, for each language resource

Tag / Measure	Precision	Recall	F-Measure
Tag 1			
Tag 2			
…			
Tag N			

Table 4 above shows the performance of the model for each output tag in each language resource. The measures are calculated from the confusion matrix for that language as shown in Table 3, and the equation details are elaborated in Chapter 3 Data Analysis section.

Table 5. Experiment Paper for the HMM Integrated Taglish POS Tagger Overall Measures

Language Resource	Weighted Mean Precision	Weighted Mean Recall	Overall Accuracy	Misclassification Rate
UD GUM (English)				
UD TRG + UD Ugnayan (Filipino)				
Annotated TweetTaglish (Taglish)				

Table 5 above presents the weighted mean precision, recall, accuracy and misclassification rate of the model, which gives an overview into how it performs for each language resource used. The statistical measures are elaborated in Chapter 3 Data Analysis.

Data Generation and Gathering Procedure

In this section we define the procedures for gathering and generating data throughout the progress of the study. This follows a typical machine learning project workflow. The procedures are divided into two (2) main phases: model training and testing phase.

Preparation and Training Phase.

1. The corpora in .conllu format will be retrieved from their sources and read into memory.
2. The corpora will be separated into 80:20 each, the training and testing sets respectively. Each sentence has an equal probability of being classified into the two sets.
3. The HMM and Maximum-entropy classifier is prepared and implemented with the Python programming language.
4. The HMM is fed the training set to learn the transition and emission probabilities.
5. The Maximum-entropy classifier learns the appropriate weights for each feature with a convex optimization algorithm which maximizes the probabilities for the labelled training set.
6. After training, the models are now ready and is saved into non-volatile memory as a file. For the HMM, the output file will simply consist of the transition and emission probabilities. The MaxEnt model will save only its calculated weights for each

feature. This is done to prevent re-training; the parameters and hyperparameters learned will simply be read from these files, when necessary.

Testing Phase.

1. After the model is trained, prepare the 20% test data segmented from the monolingual resources. Also, prepare the golden standard Taglish dataset for main testing.
2. Initialize an empty n x n confusion matrix, where n is the total number of tags the model guesses.
3. Start testing with the manually annotated Taglish dataset, by plugging each sentence into the model. For an individual sentence, the model will guess, and the output will be compared to the actual tags in the corpora. Plug the values into a confusion matrix.
4. The confusion matrix is analyzed, and the following measures are retrieved per tag:
    a. Precision
    b. Recall
    c. F-measure
5. Calculate the overall measures:
    a. Weighted Mean Precision
    b. Weighted Mean Recall
    c. Overall Accuracy
    d. Misclassification Rate
6. Repeat Step 3 and Step 5 each for the 20% English testing set, and the 20% Tagalog testing set.

7. The output of this phase will be 7 filled-out experiment papers:
    a. Confusion Matrix for the English dataset (Table 3)
    b. Confusion Matrix for the Tagalog dataset (Table 3)
    c. Confusion Matrix for the Taglish manually annotated dataset (Table 3)
    d. Performance Matrix for the English dataset (Table 4)
    e. Performance Matrix for the Tagalog dataset (Table 4)
    f. Performance Matrix for the Taglish manually annotated dataset (Table 4)
    g. Overall Performance Measure of the Model (Table 5)

Data Analysis

Learning the Hidden Markov Model parameters.

The model needs two (2) elements to its learning completion: a transition probability matrix A and an emission probability matrix B. A supervised learning algorithm will run over the three corpora for the two language resources Tagalog and English given that the words are annotated with their actual part-of-speech tags.

$$Tag = \{ Tag_1, Tag_2, Tag_3, \dots, Tag_n \}$$

$$Word = \{ Word_1, Word_2, \mathrm{Word}_3, \dots, Word_m \}$$

The tags for the model will be represented as a set of states Tag where each tag $Tag_i \; for \; 1 \leq i \leq \; n \; where \; n = |Tag|$ represents a tag in the Universal POS Tagset and the words known will be represented as a set of strings $Word = Word_1, Word_2, \dots, Word_m$ where each $Word_k \; for \; 1 \leq k \leq \; m \; where \; m = |Word|$ represents a word from an observation in the corpora.

$$A = [\, A_{11}, A_{12}, \dots, A_{1\mathrm{n}}, A_{21}, \dots, A_{\mathrm{nn}}]$$

$$where:$$

$$A_{\mathrm{ij}} = P(Tag_{\mathrm{j}}|Tag_{\mathrm{i}}) = \frac{COUNT(Tag_j \mid Tag_i)}{COUNT(Tag_i)} \; for\ 1 \leq i, j \leq n\ where\ n = |Tag|$$

Figure 16. An example transition probability matrix

Prev/Next	ADJ	ADV	INTJ	NOUN	PROPN	VERB	ADP	AUX	CCONJ	...
<s>	0.430113	0.344153	0.853258	0.678785	0.391978	0.980698	0.687689	0.044218	0.229547	...
ADJ	0.45917	0.937254	0.711287	0.906458	0.830527	0.042351	0.554345	0.916636	0.370729	...
ADV	0.98311	0.645461	0.660762	0.078954	0.264864	0.890186	0.959722	0.655161	0.276562	...
INTJ	0.768851	0.103661	0.032577	0.862244	0.019958	0.584447	0.730116	0.63626	0.528557	...
NOUN	0.203568	0.281247	0.710349	0.321764	0.905873	0.36375	0.193342	0.1475	0.272884	...
PROPN	0.907617	0.55288	0.772523	0.311312	0.805637	0.74926	0.958727	0.377794	0.859026	...
VERB	0.798779	0.01814	0.479917	0.503474	0.834907	0.587017	0.171893	0.083098	0.541199	...
ADP	0.459731	0.186495	0.301383	0.640581	0.296901	0.603656	0.417789	0.637797	0.622202	...
AUX	0.3631	0.507119	0.838607	0.158535	0.740548	0.940656	0.367669	0.782756	0.926016	...
CCONJ	0.361673	0.6894	0.802649	0.980709	0.411167	0.677871	0.974191	0.989432	0.908679	...
...	...	...	...	...	...	...	...	...	...	...

The transition probability matrix A is a matrix where each cell A_{ij} represents the probability that the model will transition from state Tag_i to state Tag_j, where both Tag_i and Tag_j is any tag in the tagset. This is structured where each row is a tag, and then the columns would be the next probable tag. Let each cell bear the probability $P(Tag_j \mid Tag_i)$. This is in alignment with the Markov property, where the next state only is fully determined from the previous state. Learning the transition probability from the corpora for each possible pair of tags is defined above, where $COUNT(Tag_i)$ is simply the number of occurrences of the previous tag Tag_i in the corpora and $COUNT(Tag_j \mid Tag_i)$ is the total number of occurrences where Tag_j appears after Tag_i.

$$B = [\, B_{11}, B_{12}, \ldots, B_{1\mathrm{m}}, B_{21}, \ldots, B_{\mathrm{nm}}]$$

$$where:$$

$$B_{\mathrm{ij}} = P(Tag_{\mathrm{i}}|Word_{\mathrm{k}}) = \frac{COUNT(Tag_i \mid Word_k)}{COUNT(Tag_i)}$$

$$for\ 1 \leq i \leq n\ where\ n = |Tag|, 1 \leq k \leq m\ where\ m = |Word|$$

Figure 17. An example emission probability matrix

Tag/Word	hello	swimming	ako	mahal	ang	...
ADJ	0.31267	0.170081656	0.893671	0.266498	0.685661	
ADV	0.138943	0.210750438	0.763147	0.140123	0.059664	
INTJ	0.783209	0.527474939	0.323782	0.560939	0.452051	
NOUN	0.758693	0.373498485	0.291127	0.183779	0.220114	
PROPN	0.045262	0.64354	0.421976	0.083797	0.518901	
VERB	0.704826	0.819287429	0.08108	0.421769	0.301106	
ADP	0.156458	0.729527284	0.607737	0.731446	0.477476	
AUP	0.486179	0.228590484	0.356819	0.074388	0.076371	
CCONJ	0.529054	0.923281445	0.868327	0.570487	0.717675	
...						

The other parameter is the emission probability matrix B where each cell B_{ik} represents the probability that a word $Word_k$ is assigned the tag Tag_i, for each tag in the tagset and for each word in the corpora. Let each cell bear the probability $P(Tag_i \mid Word_k)$. Learning the emission probability from the for each word and each tag B_{ij} is shown above , where $COUNT(Tag_i)$ is simply the occurrences of the tag Tag_i in the corpora, and $COUNT(Tag_i \mid Word_k)$ is the total number of occurrences the word $Word_k$ is assigned the tag Tag_i.

$$\pi = \pi_1, \pi_2, \ldots, \pi_n$$

$$where:$$

$$\pi_{\mathrm{i}} = P(Tag_{\mathrm{i}}|\mathrm{start}) = \frac{COUNT(Tag_i|start)}{COUNT(start)} \; for\ 1 \leq i, j \leq n\ where\ n = |Tag|$$

As defined above, the starting probability list is simply the row in the transition probability matrix where the previous state is the start state and is calculated the same. This matrix is included as the first row in the transition matrix.

Learning the Maximum Entropy Classifier parameters.

The parameters of the Maximum Entropy classifier included as an added sub-model is simply the weights that are associated with each feature. The features are indicator functions which return either 1 or 0, and are listed below:

$$f_{a\ldots b}(\mathrm{c},\mathrm{x}) = \begin{cases} 1, if\ word\ contains\ suffix\ for\ some\ tags \\ 0, otherwise \end{cases}$$

$$f_{c\ldots d}(\mathrm{c},\mathrm{x}) = \begin{cases} 1, if\ word\ contains\ prefix\ for\ some\ tags \\ 0, otherwise \end{cases}$$

$$f_{e}(\mathrm{c},\mathrm{x}) = \begin{cases} 1, if\ word\ contains\ uppercase\ and\ c == NOUN \\ 0, otherwise \end{cases}$$

$$f_{f\ldots g}(\mathrm{c},\mathrm{x}) = \begin{cases} 1, if\ word\ is\ contains\ hyphen\ for\ some\ tags \\ 0, otherwise \end{cases}$$

$$f_{h}(\mathrm{c},\mathrm{x}) = \begin{cases} 1, if\ word\ is\ all\ uppercase\ and\ c == NOUN \\ 0, otherwise \end{cases}$$

The goal of the learning phase as shown below is to find the set of weights $\widehat{w}$ that when the model is used in the training data, the total product of the probabilities is maximized. These weights determine the strength of the features for each tag. $P\left(y^{(i)}\middle|x^{(i)}\right)$ as shown below represents the probability guess of the model for the word x given the tag y.

$$\widehat{w} = argmax_{w} \prod_{i}^{M} P\left(y^{(i)}\middle|x^{(i)}\right)$$

$$where:$$

$$P\left(y^{(i)} = c\middle|x^{(i)} = x\right) = \frac{\exp\left(\sum_{i=0}^{N} w_{ci} f_i(c,x)\right)}{\sum_{c' \in C} \exp\left(\sum_{i=0}^{N} w_{c'i} f_i(c',x)\right)}$$

$$M = |Training\ Data|$$

$$N = |Features|$$

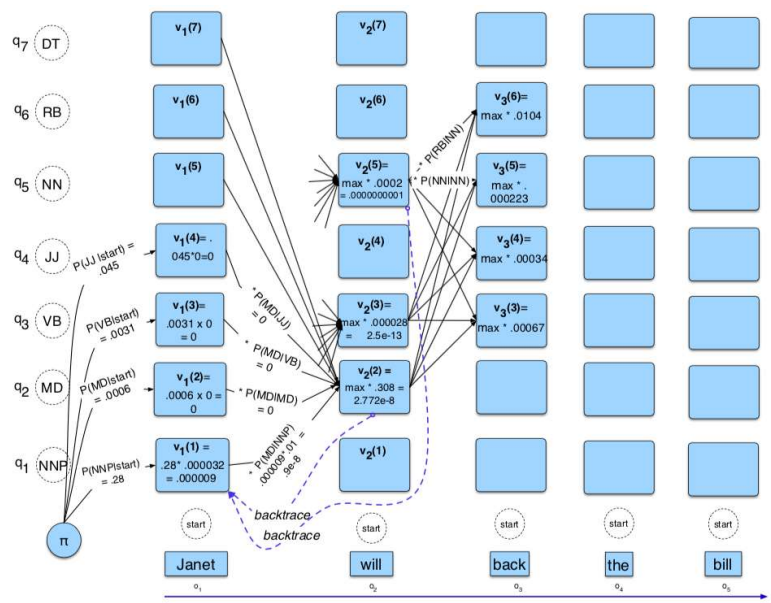
$$w = set\ of\ weights\ for\ each\ class\ and\ feature$$

$$f = set\ of\ feature\ functions, defined\ above$$

Decoding the Part-of-Speech Tags given a Sentence.

In performing guesses, the Viterbi algorithm will be used by the model. The input sentence would be a list of strings where each element is an individual token in the sentence. Let this input sentence fed as input into the model be the observation sequence $O = O_1 O_2 \dots O_n$ where each observation O_i is the ith word, and n is the number of tokens in the input sentence.

Figure 18. The Viterbi algorithm in action, and the determination of the output path in a sample input sentence "Janet will back the bill" (Jurafsky & Martin, 2021)



A trellis lattice as shown above will be constructed by the model to perform guessing, which is simply the calculation of every possible tag sequence for the words. It is represented as a matrix T where each cell T_{ij} represents the Viterbi probability of the tag Tag_i being assigned to the word O_j. The starting lattice cells are calculated by

multiplying the emission probability B_{ij} with the starting probability for each tag π_i. The subsequent lattices are calculated with regards to the maximal a-posteriori path to it with the formula $T_{ij} = MAX(T_{i,j-1} * A_{hi}) * B_{ij}$ where $T_{i,j-1}$ is the Viterbi probability of the previous word O_{j-1} for each tag Tag_i, A_{hi} is the transition probability of each tag $Tag_h\ where\ 1 \leq h \leq n, n = total\ \#.of\ tags$ (looping through all tags again in an inner loop with h as counter) to the current tag Tag_i. The maximal path $MAX(T_{i,j-1} * A_{hi})$ will only be considered for the calculation of the current lattice cell, multiplied with the emission probability B_{ij} which is the same as before.

In cases where the emission probability that the Markov model returns zero, the word is therefore not present in the corpora; this is when the word is plugged into the Maximum Entropy classifier for each tag and should return the likelihood or probability. This output is then treated as the actual emission probability of that word for that tag, and the trellis lattice calculation proceeds as normal. The calculations are defined below:

$$V_{ij} = \begin{cases} A_{0i} * B_{ij}, if\ i = 0 \\ MAX(T_{i,j-1} * A_{hi}) * B_{ij},\ \ otherwise \end{cases}$$

$$where:$$

$$B_{ij} = \mathrm{P}(\mathrm{c} = Tag_i | \mathrm{x} = [O_j, \mathrm{features_of_}O_j]) = \begin{cases} \dfrac{COUNT(Tag_i \mid O_j)}{COUNT(Tag_i)}, normally\ (HMM) \\ \dfrac{\exp(\sum_{i=0}^{N} w_{ci} f_i(c,x))}{\sum_{c' \in C} \exp(\sum_{i=0}^{N} w_{c'i} f_i(c',x))}, if\ above = 0\ (MaxEnt) \end{cases}$$

The output of the model will be a sequence $V = V_1 V_2 \ldots V_n$ otherwise known as the Viterbi path, where each V_i is the guess tag of the model for the ith word in the sentence with the maximum path probability until that word. Therefore, the final sequence will be the maximum a-posteriori path, the states of the model for the observation sequence. This

output sequence will be merged with the observation sequence to construct the final output of the program module as a list of 2-tuples as elaborated earlier in the section.

Assessing the Performance of the Model.

To identify the performance of the POS tagger, a table known as a confusion matrix will be used. A confusion matrix, otherwise known as an error matrix, is a tool used for defining the performance of classification algorithms (Kulkami and Chong, 2020). The basic example for 2-valued classification models consists of four (4) basic numbers that define the measurement metrics of the classifier. However, a POS tagger is a classifier that has more than two outputs, therefore the size of the confusion matrix depends on the number of tags that the classifier uses. The POS tagger's performance will be measured based on the F1 statistic measure of each tag, and an overall accuracy statistic. Given this confusion matrix, we will define statistical measures around the table for each tag, as elaborated in the following sections.

Table 6. A squished part-of-speech tagger confusion matrix

		PREDICTED									
	Tag	ADJ	ADV	INTJ	NOUN	PROPN	VERB	ADP	AUX	CCONJ	...
ACTUAL	ADJ	T_ADJ	.	.	.	.	.	.	.	.	.
	ADV	.	T_ADV	.	.	.	.	.	.	.	.
	INTJ	.	.	T_INTJ	.	.	.	.	.	.	.
	NOUN	.	.	.	T_NOUN	.	.	.	.	.	.
	PROPN	.	.	.	.	T_PROPN	.	.	.	.	.
	VERB	.	.	.	.	.	T_VERB	.	.	.	.
	ADP	.	.	.	.	.	.	T_ADP	.	.	.
	AUX	.	.	.	.	.	.	.	T_AUX	.	.
	CCONJ	.	.	.	.	.	.	.	.	T_CCONJ	.
	...	.	.	.	.	.	.	.	.	.	T_...

$$Conf = [Conf_{11}, Conf_{12}, \dots, Conf_{1n}, Conf_{21}, \dots, Conf_{nn}\ ]$$

Given this confusion matrix, we present the statistical measures to be employed in assessing the POS tagger that was formed in the study. This confusion matrix is filled

during testing phase, and will be the matrix $Conf$ where each cell $Conf_{rc}$ represents the number of times the word with an actual tag Tag_{r} is predicted with the tag Tag_{c}, $for\ 1 \leq r, c \leq n\ where\ n = |Tag|$. The matrix can be used to make conclusions with the performance of the model for tagging the words with their part-of-speech tags; each output class of the model is assessed with the measures as shown below. Measures for each tag are elaborated in the next sections.

$$Precision = Precision_1, Precision_2, \dots, Precision_n$$

$$Recall = Recall_1, Recall_2, \dots, Recall_n$$

$$F - Measure = F - Measure_1, F - Measure_2, \dots, F - Measure_n$$

$$where:$$

$$n = |Tag| = no.of\ output\ classes$$

The precision of a tag is simply the fraction of relevant results among the retrieved instances. In the context of the part-of-speech tagger, for some tag Tag_{i}, in all the times Tag_{i} was used, what proportion of it was correct? We count all the times Tag_{i} was both the actual and predicted tag, and where this same tag was predicted but was not the actual tag in the corpora. This is a measure of quality.

$$Precision_i = \frac{Conf_{ii}}{\sum_{r=1}^{n} Conf_{ri}}$$

Whereas the recall of a tag is simply the fraction of relevant results that were retrieved. In this same context, for some tag Tag_{i} and for all words that the actual tag was this tag, what proportion was predicted correctly against where the predicted tag was another? We count all the times Tag_{i} was both the actual and predicted tag, and where this tag was the actual tag but not predicted correctly. This is a measure of quantity.

$$Recall_i = \frac{Conf_{ii}}{\sum_{c=1}^{n} Conf_{ic}}$$

The F-Measure of a tag $F1_i$ states the harmonious equilibrium between the precision and recall of that tag. This is the measure for the classifier's usefulness with that tag scenario. The measure lies as a fraction between 0 and 1, where 1 denotes perfect precision and recall, whereas 0 being the opposite.

$$F - Measure_i = 2 \cdot \frac{(Precision_i)(Recall_i)}{Precision_i + Recall_i}$$

Additional statistical measures are utilized towards calculating the overall assessment of the model. These measures are the weighted mean precision, weighted mean recall, overall accuracy and the misclassification rate of the model, respectively.

$$Weighted\ Mean\ Precision = \frac{\sum_{i=1}^{n}(COUNT(Tag_i) * Precision_i)}{no.of\ words}$$

$$Weighted\ Mean\ Recall = \frac{\sum_{i=1}^{n}(COUNT(Tag_i) * Recall_i)}{no.of\ words}$$

$$Overall\ Accuracy = \frac{no.\ of\ correctly\ tagged\ words}{no.\ of\ words}$$

$$Misclassification\ Rate = 1 - \text{Overall Accuracy}$$

The weighted mean precision and recall of the model is simply the sum of all precision/recall measures after multiplying them with their respective class proportions. The overall accuracy of the model counts the ratio of the number of correct tags to the total number of tags. The higher the percentage, the more accurate the model is as per the testing details. The misclassification rate is simply 1 minus the accuracy. These

combined measures will be calculated for each of the language resources to answer SOPs 1-3.

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APPENDICES

APPENDIX A

Experiment Paper #1.

Confusion Matrix (per Language)

		PREDICTED							
		ADJ	ADV	AUX	CCONJ	DET	INTJ	NOUN	...
ACTUAL	ADJ								
	ADV								
	AUX								
	CCONJ								
	DET								
	INTJ								
	NOUN								
	...								

Experiment Paper #2.

Performance Matrix (per Language)

Tag	Precision	Recall	F-Measure
ADJ			
ADV			
AUX			
CCONJ			
DET			
INTJ			
NOUN			
...			

Experiment Paper #3.

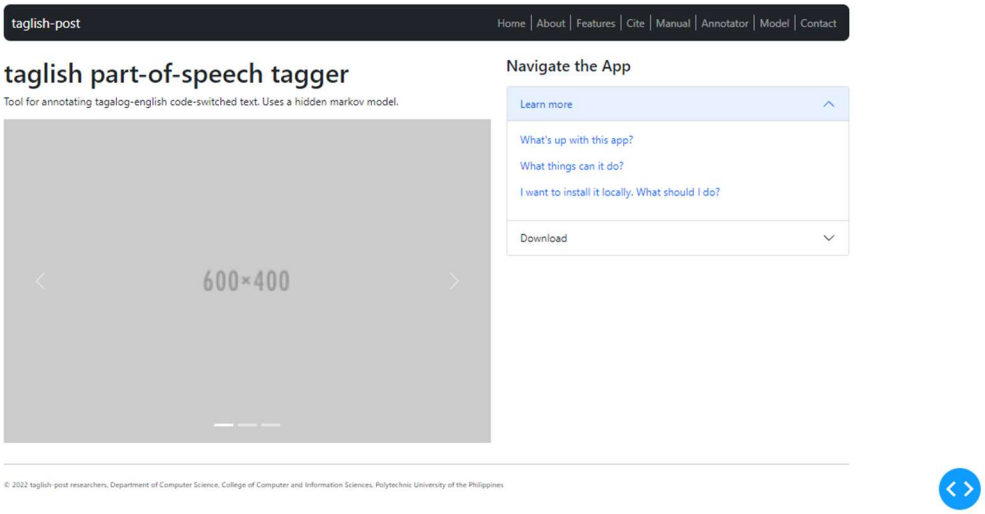
Overall Performance Matrix

Language Resource	Weighted Mean Precision	Weighted Mean Recall	Overall Accuracy	Misclassification Rate
English				
Tagalog				
Taglish				

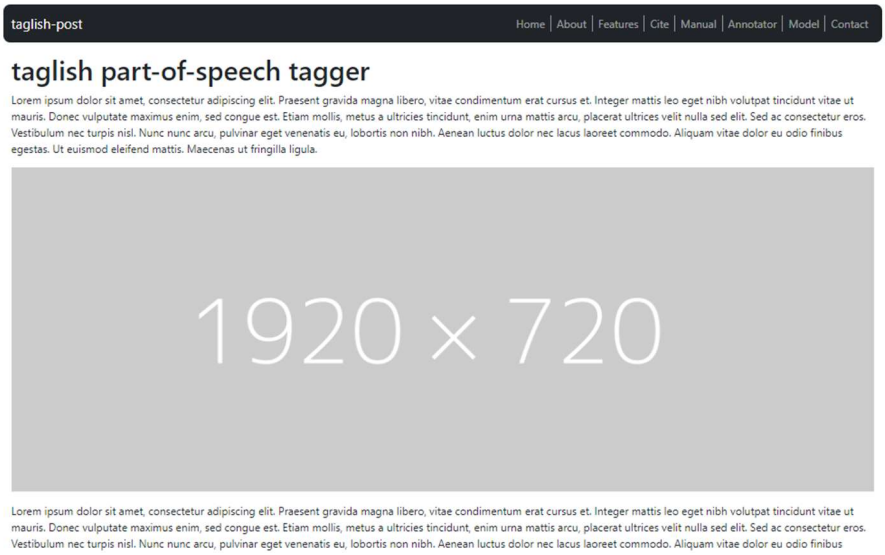
APPENDIX B

Screen Mockup

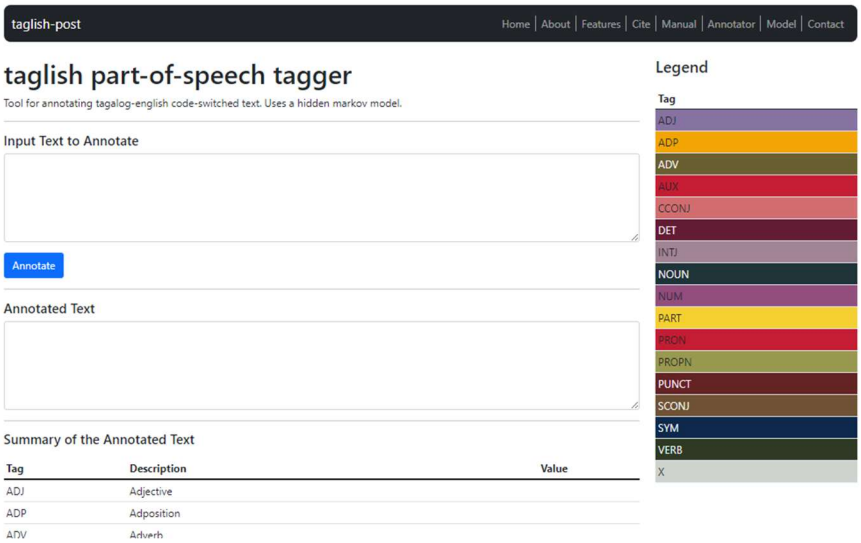
Home Page



About Page



Model Page



Model Page (Summary of Annotated)

Summary of the Annotated Text

Tag	Description	Value
ADJ	Adjective	0
ADP	Adposition	0
ADV	Adverb	0
AUX	Auxiliary	0
CCONJ	Coordinating Conjunction	0
DET	Determiner	0
INTJ	Interjection	0
NOUN	Noun	0
NUM	Numeral	0
PART	Particle	0
PRON	Pronoun	0
PROPN	Proper Noun	0
PUNCT	Punctuation	0
SCONJ	Subordinating Conjunction	0
SYM	Symbol	0
VERB	Verb	0
X	Other	0
Total Words		0
Total Tokens		0
Unique Tokens		0
Unique Parts of Speech		0

SYM
VERB
X

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